

Week 7: HealthConnect Analytics Testing & Validation Report

Track: Data Analytics Track Project: HealthConnect Clinic Experience Lab Status: Validated via Python (Pandas)

1. Programmatic Terminal Validation Logs

```
(venv) danielmoi820@penguin:~$ python3 data.py Dataset successfully loaded. Total rows: 5000, Total columns: 18 --- Column Data Types & Missing Values --- RangeIndex: 5000 entries, 0 to 4999 Data columns (total 18 columns): 0 appointment_id 5000 non-null str 1 patient_id 5000 non-null str 2 gender 5000 non-null str 3 age 5000 non-null int64 4 age_group 5000 non-null str 5 appointment_type 5000 non-null str 6 booking_date 5000 non-null str 7 appointment_date 5000 non-null str 8 appointment_day 5000 non-null str 9 appointment_time 5000 non-null str 10 booking_lead_days 5000 non-null int64 11 previous_appointments 5000 non-null int64 12 previous_no_shows 5000 non-null int64 13 reminder_sent 5000 non-null str 14 reminder_channel 3634 non-null str (1,366 missing) 15 distance_to_clinic_km 4910 non-null float64 (90 missing) 16 waiting_time_minutes 4940 non-null float64 (60 missing) 17 appointment_outcome 5000 non-null str Memory usage: 703.3 KB
```

2. Testing & Validation Record Table

Component Tested	Dashboard KPI Summary Cards & Demographic Visualisations
Testing Objective	Verify whether dashboard metric calculations and visual trends agree with the underlying raw dataset (HealthConnect_Appointment_Data.csv).
Test / Scenario	Programmatically recalculating overall no-show rates and segment breakdowns using Python (Pandas) to cross-check dashboard values.
Expected Result	Dashboard KPI cards and filter outputs should match pythonically computed statistics exactly across all 5,000 records.
Actual Result	Core no-show rates matched successfully; data quality audit confirmed 5,000 rows, 18 columns, and identified exact missing values in optional fields.
Pass / Fail Status	Pass (with data quality notes)
Issue Identified	Missing entries detected in reminder_channel (1,366), distance_to_clinic_km (90), and waiting_time_minutes (60).
Action Taken	Documented exact missing value counts from terminal logs, verified outcome distribution robustness, and updated dashboard layout notes.
Retest Result	Re-ran verification scripts confirming data integrity and memory allocation (703.3 KB) across filtered segments.
Collaborating Track	Data Science / Project Management
Evidence	Python terminal execution logs, missing value counts, and verified dataset structural metadata.

Audit Summary: Programmatic validation confirms that the primary appointment outcome and demographic variables are 100% complete, providing a reliable foundation for downstream Data Science modeling and dashboard deployment.